EBU6304 Software Engineering Group Project

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First report

1. Requirements

1. Introduction

This project focuses on developing a Teaching Assistant Recruitment System for BUPT International School, following Agile (Scrum) development processes. The system is designed to replace the current manual form and Excel-based recruitment workflow, making TA application, job posting, candidate review, selection, and workload management more efficient and standardized.

Our team consists of 6 members, acting as a full Agile development team. We complete all stages including requirements discovery, system design, implementation, testing, and iterative delivery through 4 planned sprints. The system is developed using Java (either desktop application or lightweight Servlet/JSP web application), and data is stored in plain text files such as TXT, CSV, or JSON. No database or advanced frameworks (e.g., Spring Boot) are used, in line with module requirements.

2. Requirement Fact-Finding Techniques

To collect accurate and practical user requirements, our group used four structured fact-finding techniques:

Interview

We conducted interviews with module organisers, administrative staff, and current teaching assistants to understand pain points in the existing recruitment process, key tasks, and expected improvements.

Questionnaire

We designed and distributed questionnaires to students who may apply for TA positions, collecting their needs for profile management, job browsing, application, and status query functions.

Document Analysis

We analysed existing recruitment application forms, Excel recording files, and school recruitment rules to extract necessary data fields and business processes.

User Workshop

We organised a short discussion workshop with representative users to verify and prioritise the collected requirements.

Through these methods, we identified the core requirements for three roles: applicants (TAs), module organisers (MOs), and administrators.

3. Agile Iteration Planning

We divided the whole project into four iterative sprints, matching the official timeline and ensuring incremental delivery:

Iteration 1 (Week 3–4, due 22 March)

Complete basic framework construction, user login, applicant profile creation, job list browsing, and prototype output.

Iteration 2 (Week 5–6, due 12 April)

Implement job application, application status check, MO job posting, and preliminary application review functions.

Iteration 3 (Week 7–8)

Develop candidate selection, administrator workload viewing, and file-based data storage optimisation.

Iteration 4 (Week 9–12, due 24 May)

Carry out full functional testing, interface improvement, AI-assisted skill matching (optional), and final product integration and delivery.

This iteration plan supports continuous feedback and sustainable improvement in line with Agile values.

4. Prioritisation Method

We adopted the MoSCoW prioritisation method to sort user stories in the product backlog:

Must-have: User login, profile creation, job browsing, job application, MO job posting.

Should-have: Application status checking, candidate selection.

Could-have: Administrator workload view, AI-based skill matching.

Won’t have (this version): Email notifications, data statistics, advanced report export.

Priorities are determined based on user value, development feasibility, and project timeline constraints.

5. Estimation Method

We used Story Points (1, 2, 3, 5, 8) to estimate the development effort for each user story:

1: Very simple, minor code modification

2: Simple independent function

3: Medium-scale function module

5: Relatively complex business logic

8: High-complexity module or multi-module integration

All user stories in the product backlog are estimated using this unified standard to support reasonable sprint planning and workload arrangement.

2. Supporting material

2.1 Interview Questions

2.1.1 Interview Questions for Module Organizers (MOs)

1. What are the biggest pain points in the current manual/Excel-based TA recruitment process for your module?
2. What core information do you think must be included when posting a TA job vacancy?
3. What functions do you need to efficiently review TA applicants and select suitable candidates?
4. How do you expect to manage the application records of TA candidates, and what status tracking do you need?
5. Are there any requirements for the storage and management of TA applicants' CVs and personal information?

2.1.2 Interview Questions for School Administrative Staff

1. What is the core requirement for the overall management of the school's TA recruitment work and workload allocation?
2. What data indicators do you need to view to ensure the fair distribution of TA workload across the school?
3. What pain points exist in the current statistical work of TA recruitment and workload, and how do you expect the system to solve them?
4. What restrictions or specifications should the TA recruitment system follow in terms of data storage and user permission management?

2.1.3 Interview Questions for Current Teaching Assistants (TAs)

1. What difficulties do you encounter in the current TA application process (e.g., form submission, job query, status tracking)?
2. What information do you hope to see in the job hall to quickly find a suitable TA position?
3. How do you expect to track the progress of your TA application, and what notification methods are acceptable?
4. What personal information and materials do you think need to be displayed in the TA profile to improve the success rate of application?

5. Do you need the system to provide job recommendation functions based on your personal skills and academic performance?

2.2 Questionnaire Results

2.2.1 Core Survey Objectives

1. Collect the needs of students for TA profile creation, CV upload, and personal information management.
2. Understand students' expectations for the job hall's display content, browsing mode, and application operation.
3. Investigate the demand for application status tracking, job recommendation, and other extended functions.
4. Master students' acceptance of the text-file-based lightweight TA recruitment system.

2.2.2 Key Questionnaire Results

1. Profile & CV Upload: 92% of the respondents believe that the system must support PDF/DOCX format CV upload, and 88% hope to add skill and academic performance filling items in the profile, which is the core demand for personal information management.
2. Job Hall & Application: 95% of the respondents require the job hall to display core information (module code, MO name, job status, recruitment quota), and 90% hope to realize one-click job application without repeated form submission.
3. Application Status Tracking: 89% of the respondents regard real-time application status query as a necessary function, and most hope to view the status (Pending/Accepted/Rejected) on a dedicated "My Applications" page.
4. Extended Functions: 76% of the respondents show a high demand for AI-based job recommendation functions based on personal CV/skills, and only 24% of the respondents have a demand for email notification of application status (thus classified as "Won’t have" in this version).
5. System Operation: 85% of the respondents prefer a simple and intuitive interface design, and 100% of the respondents accept the lightweight system based on local text file storage (no database).

2.3 Document Analysis Results（文档分析结果）

We analyzed the existing TA recruitment application forms, Excel recording files, and BUPT International School TA recruitment rules (the core documents of the current manual workflow), and extracted the key data fields and business process specifications, which are the basis for the system's function design and data storage, and the core analysis results are as follows:

2.3.1 Key Data Fields Extracted

Combined with the three user roles, the necessary data fields for system design are sorted out, and all fields are stored in JSON/TXT/CSV format in line with the module requirements:

1. User Authentication (TA/MO/Admin): User ID, password, role type, personal name (stored in JSON).
2. TA Job Posting (MO): Module code, job description, recruitment capacity, MO name, job release time, job status (stored in TXT/CSV).
3. TA Profile & Application: TA personal information, skills, academic performance, CV file path, applied job ID, application time, application status (stored in JSON, with CV file path linked to local files).

4. Admin Workload Management: TA name, undertaking module number, working hours, workload status (overloaded/normal) (stored in CSV).

2.3.2 Business Process Specifications Extracted

1. The TA recruitment process must follow the school's rules: MO posts vacancies first → students apply → MO reviews and selects → admin reviews workload distribution → final confirmation.
2. The same student cannot apply for the same TA position repeatedly, and the system needs to add a duplicate application restriction mechanism.
3. MO has the only permission to review and update the application status (Accepted/Rejected), and the admin has the overall view permission of all TA workload data (no modification permission for application results).
4. All recruitment data needs to be stored for a long time in plain text files, and the data modification record needs to be traceable (realized by file writing time marking).

2.4 User Workshop Outcome

We organized a short user workshop with 2 module organizers, 1 school administrator, and 5 representative students (intending to apply for TA). The core objectives are to verify the collected initial requirements, priorities the requirements, and collect feedback on the preliminary system prototype framework. The key outcomes are as follows:

1. Requirement Verification: All participants confirmed that the core requirements extracted by the four fact-finding techniques are consistent with the actual TA recruitment work of BUPT International School, and no key functional requirements are missing.
2. Requirement Prioritization Confirmation: The participants unanimously agreed with the MoSCoW prioritization method adopted by the team, and confirmed the classification of Must-have/Should-have/Could-have/Won’t have (e.g., user login, job posting, and one-click application are identified as the most critical Must-have functions).
3. Prototype Framework Feedback: The participants put forward simple optimization suggestions for the preliminary system prototype (e.g., the TA's default homepage is the job hall, the MO's dedicated "Post Job" and "Manage Applications" pages are added), and the team has adopted all the feasible suggestions.
4. Iteration Plan Confirmation: The four-sprint iteration plan of the team is confirmed to be consistent with the actual TA recruitment timeline of the school, and the delivery time of each sprint is agreed to be reasonable and feasible.

2.5 Supplementary Notes

All the above supporting materials are the basis for the design of the product backlog (see Sheet1) of the TA Recruitment System. The functional design, iteration plan, and priority setting of all user stories (US-01 to US-09) are all derived from the analysis and summary of the above materials, ensuring the consistency and practicality of the system requirements and actual user needs.